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main

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Metadata and Protein Data for Mo...

README.md

bar_graph.png

main.py

patient.py

scatter_plot.png

BME2315_Module01 / main.py

LiamSunshine Saving graphs

94501d2 · 8 minutes ago

History

CodeBlame48 lines (38 loc) · 1.52 KB

```
1  from patient import Patient
2  import numpy as np
3  import matplotlib.pyplot as plt
4
5  # Create Patient objects using the data from the CSV file
6  Patient.instantiate_from_csv("Metadata and Protein Data for Module 1.csv")
7
8  # Sort patients by age at death
9  sorted_patients = sorted(Patient.all_patients, key=lambda patient: patient.age_at_death)
10
11 # Print the sorted patients
12 for patient in sorted_patients:
13     print(patient)
14
15 # Filter and print female patients with dementia
16 filtered_patients = Patient.filter(
17     Patient.all_patients,
18     sex="Female",
19     cognitive_status="Dementia"
20 )
21
22 for patient in filtered_patients:
23     print(patient)
24
25 # Compare mean ABeta42 levels between female and male patients
26 female_abeta42 = [patient.abeta42 for patient in Patient.all_patients if patient.sex == "Female"]
27 male_abeta42 = [patient.abeta42 for patient in Patient.all_patients if patient.sex == "Male"]
28
29 means = [np.mean(female_abeta42), np.mean(male_abeta42)]
30 standard_deviations = [np.std(female_abeta42), np.std(male_abeta42)]
31
32 plt.bar(["Female", "Male"], means, yerr=standard_deviations, capsize=5)
33 plt.xlabel("Sex")
34 plt.ylabel("ABeta42 (pg/ug)")
35 plt.title("Mean ABeta42 Levels by Sex")
36 plt.savefig("bar_graph.png")
37 plt.show()
38
39 # Compare ABeta40 and ABeta42 levels
40 abeta40 = [patient.abeta40 for patient in Patient.all_patients]
41 abeta42 = [patient.abeta42 for patient in Patient.all_patients]
42
43 plt.scatter(abeta40, abeta42)
44 plt.xlabel("ABeta40 (pg/ug)")
45 plt.ylabel("ABeta42 (pg/ug)")
46 plt.title("ABeta40 vs. ABeta42 Levels")
47 plt.savefig("scatter_plot.png")
48 plt.show()
```